

DNS simulations of inertial flow phenomena in 3D intersecting rough fractures: Impact of roughness and intersection angle on non-linear flow regimes

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ABSTRACT

Understanding fluid flow within fractured rock masses is critical for a wide range of applications, including groundwater modeling, geothermal energy extraction, mine water management, and resource recovery from fractured reservoirs. While flow in single fractures has been extensively studied, the complex interactions introduced by intersecting fractures with varying roughness and aperture significantly influence inertial flow behavior and transport dynamics.

This study applies three-dimensional Direct Numerical Simulations (DNS) using scanned rock fracture surfaces to quantify how roughness, aperture, and intersection angle affect non-Darcy flow regimes. Fluid flow through X-shaped, rough-walled fracture intersections was simulated over a range of flow rates (0–200 mL/s) revealing that the transition to inertial flow occurred at Reynolds numbers between 10 and 100, depending on geometry. The results show that fracture roughness is the primary driver of non-linear flow behavior, lowering the critical hydraulic gradient by up to 60 % and increasing the Forchheimer coefficient β by up to an order of magnitude.

Non-linear flow behavior was characterized using the Forchheimer and Izbash equations, with the Izbash exponent n ranging from 1.2 to 1.8, depending on roughness and angle. The inertial coefficients (β and n) increase with roughness and intersection angle, reflecting enhanced inertial resistance. Empirical power-law relationships were derived to predict these coefficients such as $\beta \propto K^{-5/3}$.

These findings provide new insights into fluid dynamics in complex fracture networks and offer predictive tools for permeability modeling in subsurface systems. The results are relevant to geothermal reservoir engineering, mine dewatering, carbon storage, and other fluid-driven subsurface technologies.

1. Introduction

Accurately modeling fluid flow through fractured rock masses is essential for improving geothermal energy extraction (Mu et al., 2025), carbon dioxide sequestration (Xu et al., 2024), hydrocarbon recovery (Rodríguez et al., 2024), underground oil storage (Wang et al., 2015), and hydraulic fracturing (Aghababaei et al., 2024). Flow through natural fracture networks is governed by roughness, aperture, and connectivity, which significantly complicate predictive models, particularly under non-Darcy conditions.

Most early investigations focused on flow within single fractures, using the cubic law under laminar conditions and the Forchheimer equation for higher Reynolds numbers (Javadi et al., 2010; Zimmerman et al., 2004). These studies established that fracture surface roughness induces nonlinearities even at low Reynolds numbers (Xiong et al., 2020), challenging the assumption that flow in fractured media is predominantly laminar. (Zhou et al., 2019) advanced this understanding by identifying a universal relationship between viscous and inertial permeability in geologic porous media, showing that inertial effects can be predicted from Darcy-scale properties.

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Table 1
Summary of previous studies on fluid flow in fractured media.

Study	Fracture Geometry	Dimension	Key Focus
(Johnson et al., 2006)	Intersections	2D	Investigating how surface roughness in fracture intersections enhances fluid mixing.
(Zou et al., 2017)	90 degrees X-shape	3D	Experimentally and numerically investigate how inertial flow affects solute mixing and partitioning at rough-walled fracture intersections.
(Zhang et al., 2021)	Y-shape	2D	Numerically analyze how bifurcation angle, surface roughness, and flow direction affect forward and reverse flow behaviors in rough-walled Y fractures.
(Li et al., 2022)	Y-shape	2D	Investigate how geometric properties, roughness, intersecting angle, and aperture, combined affect nonlinear flow behavior.
(Liu et al., 2022)	X and Y-shapes	3D	Develop and validate a generalized nonlinear flow model for 3D rock fracture intersections, quantifying how geometric factors, intersecting angle, aperture ratio, and roughness affect inertial flow behavior.
(Zhang and Li, 2024)	Single fracture	3D	Develop a nonlinear flow model for rough fractures with geometric heterogeneity by improving Izbash's law, allowing accurate simulation of inertial flow using measurable geometric parameters.
(Han et al., 2025)	Intersections	3D	Analyze how fracture roughness, aperture, scale, and intersection angle affect the transition from linear to nonlinear flow in crossed rock fractures and develop a predictive model for Forchheimer parameters.

Recent studies, including Zhou et al. (2020), highlight how wall slip conditions further complicate this picture. Slip boundary conditions, arising from hydrophobic surface chemistry or thin wetting films, enhance or reduce permeability depending on the balance between increased velocity and eddy formation. Their direct numerical simulations showed that at high pressure gradients, eddy growth counteracts the positive effects of slip, leading to an eventual reduction in permeability.

Previous studies have extensively explored the influence of single-fracture roughness on fluid flow and solute transport (Luo et al., 2022; Shi et al., 2024; Xing et al., 2021; Yang et al., 2025; Zhai et al., 2025; Zhang et al., 2025, 2024). However, the complexity of natural fracture networks shaped by interactions, intersecting angles, aperture variations, and roughness poses challenges that remain insufficiently addressed. Intersections cause fluid convergence, deflection, and redistribution, creating complex local flow regimes. Wang et al. (2022) proposed a nonlinear flow model that quantifies the interaction between intersecting fracture branches using coefficient matrices to account for both linear and nonlinear effects. Their model captures direction-dependent behavior and provides estimates for critical Reynolds numbers that demarcate the transition from laminar to inertial flow, which were found to range from 51 to 105 in 2D intersections.

Extending this analysis into 3D, Karimzade et al. (2023) explored the hydraulic properties of crossed rough-walled fractures using Navier–Stokes-based simulations. They demonstrated that factors such as intersection angle, surface roughness, and intersection length significantly affect transmissivity. Particularly, they showed that rougher intersections yield greater fluctuations in effective transmissivity, and that

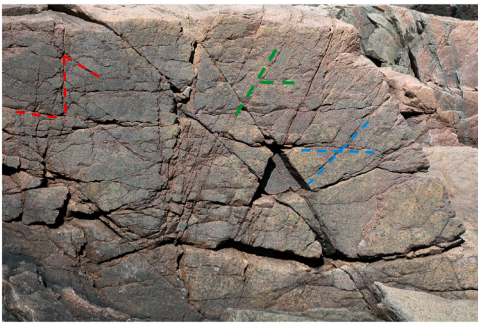


Fig. 1. Typical fracture patterns observed in granite outcrops Cadillac Mountain (Mount Desert Island, Maine, USA) highlighting natural occurrences of X-, Y-, and Z-shaped fracture intersections. (James, 2006).

intersection effects diminish only at sufficiently large scales relative to the representative elementary volume (REV).

Table 1 provides a summary of key previous studies that have addressed fluid flow in fractured media. Although substantial progress has been made in modeling nonlinear flow in both single and intersecting fractures, most studies still emphasize 2D or idealized settings. The effects of roughness and inertia in 3D intersecting rough fractures remain insufficiently explored.

To address this gap, we conduct Direct Numerical Simulations (DNS) of inertial flow through 3D X-shaped, the most common configuration observed in fractured rock outcrops (Fig. 1), fractures with controlled roughness and aperture. Our 3D DNS simulations address a critical gap in modeling non-Darcy flow in realistic fracture geometries, directly informing flow predictions for CO₂ storage, geothermal, and oil and reservoir management.

X-intersections were chosen due to their geometric symmetry and their ability to represent complex fracture connectivity scenarios more realistically than simpler Y- or Z-type intersections. By analyzing X-fractures, this study aims to isolate the impact of key geometric parameters roughness, intersection angle, and aperture on eddy development, pressure gradients, and permeability changes under inertial flow conditions.

While the scope of this study is intentionally focused on X-shaped fracture intersections under symmetric boundary conditions. Other configurations such as Y- and Z-shaped intersections, as well as non-symmetric boundary conditions, may exhibit different flow dynamics and scaling behavior. These scenarios are not addressed here, the findings presented here serve as a foundation upon which such future work can build.

To explore these phenomena, we developed twenty-four high-fidelity 3D models using real rock scans to capture realistic surface roughness. Through numerical simulations, we examine how inertial forces, and geometrical variations influence flow constrictions, eddy zones, and pressure distribution across intersecting fractures.

To address the challenges of modeling non-Darcy flow in complex fractured media, this study aims to quantify the influence of inertial effects on fluid flow patterns and pressure gradients in three-dimensional intersecting rough fractures. It further investigates how fracture roughness governs the transition from laminar to inertial flow regimes and evaluates the impact of intersection angles and aperture variations on eddy formation and flow constriction. The study also seeks to develop predictive correlations that relate key geometric parameters, including roughness, angle, and aperture, to the onset and behavior of inertial flow phenomena in rough-walled, intersecting fracture systems. To address these questions, dimensionless numbers including the Forchheimer β coefficient and the Izbash exponent (n) are employed to characterize non-linear flow behavior. Inertial flow arises in rough-walled fractures when fluid velocity increases to the point that viscous forces alone can no longer account for the observed pressure gradients.

The Forchheimer equation governs transitional inertial flows ($Re \sim 10$ – 100) and introduces a quadratic velocity term to model energy losses due to convective acceleration (Barree and Conway, 2004), streamline curvature, flow separation, and localized recirculation (Cunningham et al., 2020), even under nominally laminar conditions (Li et al., 2021). In contrast, the Izbash equation applies to turbulent regimes ($Re > 100$) (Singh et al., 2024) and adopts a power-law form, where the exponent (n) quantifies the degree of non-linearity influenced by fracture roughness and geometric complexity (Zhang et al., 2022; Zhang and Li, 2024).

Both models have been validated through experimental and numerical studies, especially within complex fracture networks (Chen et al., 2015; Huang and Ayoub, 2008; Wang et al., 2022; Zhang et al., 2022; Zhang and Li, 2024). Crucially, the onset of non-Darcy flow is not governed solely by flow rate (Cunningham et al., 2020), it is highly sensitive to geometric features such as roughness, contact area, and intersection angle (Karimzade et al., 2023; Kim and Yeo, 2025; Yang et al., 2021; Yunjin et al., 2005; Zhang et al., 2024). This underscores the need for predictive models that explicitly link fracture geometry to inertial thresholds and pressure losses. In addition, stress fields and fluid pressure can dynamically alter fracture apertures through dilation or closure (Sisavath et al., 2003).

Reynolds number (Re) and hydraulic gradient (i) are used to define the transition between linear and non-linear flow regimes. Finally, this study proposes empirical relationships to estimate permeability in X-fracture systems, thereby contributing to improved reservoir modeling.

2. Methods

This study examines inertial flow transitions in 3D rough-walled X-shaped fracture intersections. Synthetic fracture models were derived from high-resolution scans of laboratory-generated fractures with controlled roughness, aperture a , and intersection angle θ .

2.1. Darcy and non-Darcy flow representation

Flow behavior was analyzed in terms of the hydraulic gradient i and specific discharge $q = Q/A$, with effective hydraulic conductivity computed as $K = q/i$. Two well-established empirical models, the Forchheimer and Izbash equations, were fitted to the simulation results to characterize deviations from Darcy's law due to inertial effects. The Forchheimer equation is particularly effective in capturing transitional inertial flow regimes ($Re \approx 10$ – 100), where both viscous and quadratic inertial losses are significant. On the other hand, the Izbash equation is suited for describing fully inertial or turbulent flow conditions, providing insight into the degree of non-linearity in flow behavior. These two models complement each other by encompassing distinct flow regimes relevant to rough-walled fracture systems. Although other non-Darcy models exist, the combined use of Forchheimer and Izbash equations offers a practical and well-established framework to investigate the transition from linear to nonlinear flow under the experimental conditions examined in this work.

2.1.1. Forchheimer equation in pressure and gradient form

The pressure-based Forchheimer equation is expressed as:

$$-\nabla p = \frac{\mu}{k_v} v + \frac{\rho}{k_i} v^2 \quad (1)$$

where:

- ∇p : pressure gradient [Pa/m]
- v : superficial velocity (specific discharge) [m/s]
- k_v : viscous permeability [m^2]
- k_i : inertial permeability [m^2]
- μ : dynamic viscosity [Pa·s]
- ρ : fluid density [kg/m^3]

Dividing Eq. (1) by ρg , we obtain the hydraulic gradient form:

$$i = \frac{q}{K} + \frac{\beta}{g} q^2 \quad (2)$$

where:

- i : dimensionless hydraulic gradient
- $K = \frac{k_v \rho g}{\mu}$: intrinsic hydraulic conductivity [m/s]
- β : inertial resistance coefficient [1/m]

Thus, Eq. (2) captures both viscous and inertial losses. The quadratic coefficient b from simulation fits corresponds to β/g , allowing back-calculation of β .

2.1.2. Izbash power-law model for non-Darcy flow

To isolate strictly non-linear flow behavior, we also fitted the Izbash equation in the form:

$$i = c \cdot q^n \quad (3)$$

where:

- c : empirical coefficient [units vary]
- n : nonlinearity exponent (typically $1 < n < 2$)

The Izbash model does not resolve the Darcy regime but is suited for capturing fully inertial or transitional flow behavior. When plotted as $K = q/i$ versus i , the exponent n governs the curvature of the resulting conductivity trend.

2.1.3. Curve fitting procedure and parameter estimation

The i - q data extracted from Fluent simulations were fitted using nonlinear least squares regression in MATLAB. For Forchheimer fits, the form $i = \alpha \cdot q + b \cdot q^2$ was used, where:

$$\alpha = \frac{1}{K} \text{ and } \beta = b \cdot g \quad (4)$$

The coefficient α was fixed based on the average of the first three simulation points in the low-velocity (Darcy) regime. The inertial coefficient b was then estimated using the remaining data. In this formulation, β is obtained via multiplication with gravitational acceleration $g = 9.81 m/s^2$.

This practice is to isolate the inertial contribution captured by b , reducing sensitivity to the fitting range in the laminar regime. The resulting fits yielded R^2 values greater than 0.95 across all cases, indicating that the Forchheimer quadratic model provides a strong approximation of the observed non-linear behavior in our Reynolds number range.

For Izbash fits, we applied Eq. (3) with an Initial guess of the non-linear fitting algorithm and do not imply an assumption of Darcy flow. $c = 1$, $n = 1$ were used, and bounds were not explicitly imposed. Fitting was performed per angle (θ), aperture a , and roughness level.

2.1.4. Post-Processing and data presentation

Although fitting was performed in the i - q space, results were visualized in K - i coordinates to emphasize effective conductivity across flow regimes. Conductivity values were calculated from fitted curves using:

$$K(i) = \frac{q(i)}{i} \quad (5)$$

for each model, and results were compared across geometric and roughness conditions. Coefficients β , c , and n were further analyzed for correlations with hydraulic conductivity and fracture geometry.

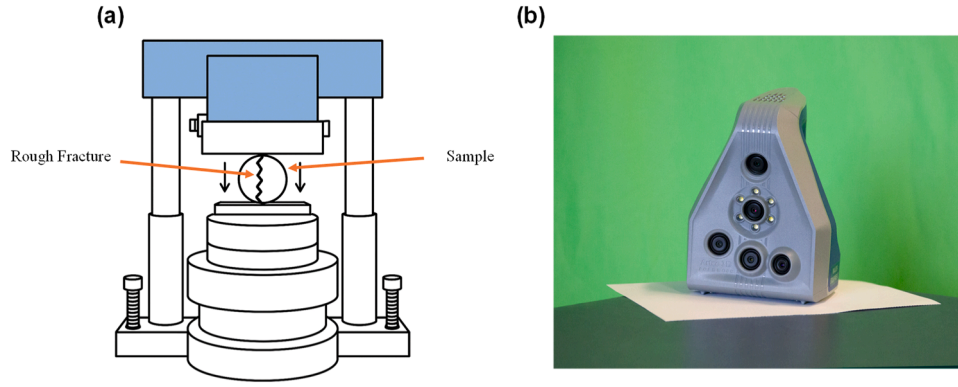


Fig. 2. Splitting process (a) and the Artec Space Spider 3D scanner (b) used in the study.

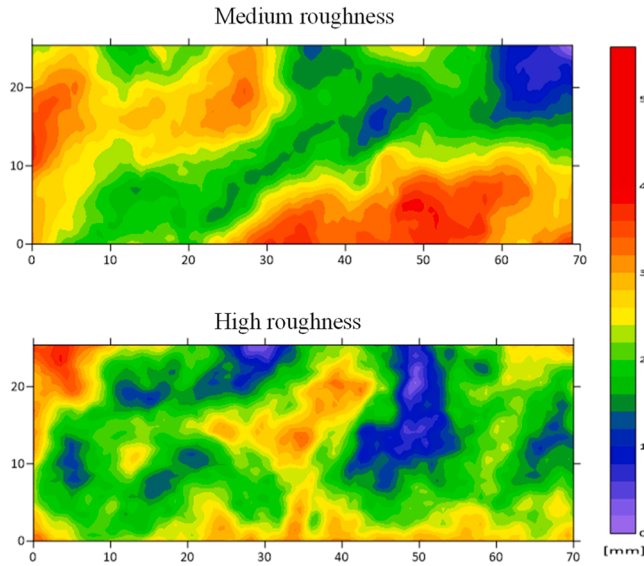


Fig. 3. Scanned rough fracture surfaces: high and medium.

2.2. Fracture geometry and model construction

2.2.1. Sample preparation and surface scanning

The proposed approach combines different scanned fracture surfaces of individual mechanically created fractures, modeling of synthetic fractures, and computational fluid dynamics simulations. All specimens were drilled perpendicular to the bedding to mitigate the influence of bedding and regulate the splitting process. In the laboratory, the rough rock fractures were generated by dividing the intact cylindrical rock samples into two halves (Fig. 2a) through the application of a compressive uniaxial testing machine at a rate of 0.1 psi/sec, resulting in the formation of mode I tensile fractures. To ensure a variety of fracture roughness characteristics, the samples were collected from different geological formations across North Dakota, each exhibiting distinct grain sizes, the fine-grained samples were sourced from the Bakken Formation, while the coarser-grained samples were obtained from the Deadwood formation.

To accurately reproduce the real fracture characteristics, the rock fracture surface was scanned using a noncontact three-dimensional optical high-resolution scanner (Artec Space Spider, Fig. 2b).

The measurement accuracy of this device is 0.05 mm for the 3D point accuracy, 0.1 mm for the 3D resolution, and 0.05 mm+0.3 mm/m for the 3D accuracy over distance. The two fracture surfaces were then digitized as shown in (Fig. 3). The numerical rock fracture surfaces were used as the primary surfaces to generate the 3D numerical fracture

Table 2

Z₂ standard deviation and mean surface JRC for each roughness category.

Surface	Mean Z ₂	Z ₂ Standard Deviation	Mean JRC
High Roughness	0.25	0.033	10
Medium Roughness	0.17	0.008	5
Parallel Plate Model	0	0	0

models.

2.2.2. Roughness quantification

For each scanned 3D fracture surface, we quantified roughness by applying a 3D→2D profile-based approach. Specifically, 200 evenly spaced profiles were extracted in the flow-parallel direction from the digital surface model. For each profile, we computed the Z₂ (Eq (6)) roughness statistic. The JRC for a given surface was obtained by converting each profile's Z₂ value to JRC via the Li and Zhang (2015) correlation (Eq (6)), obtained from Barton and Choubey (1977) and then averaging across all profiles.

$$JRC = 98.718 \cdot Z_2^{1.6833} \quad (6)$$

Z₂ captures the 2D average slope change and is given by:

$$Z_2 = \left[\frac{1}{N_t} \sum \left(\frac{Z_{i-1} - Z_i}{x_{i-1} - x_i} \right)^2 \right]^{\frac{1}{2}} \quad (7)$$

Z_i and x_i are the elevation and horizontal coordinates of the fracture profile points. N_t is the total number of measurements over the profile. This formulation quantifies the local slope variations of the surface and provides a statistical basis for estimating JRC.

A MATLAB code was used to estimate both the statistical roughness parameter Z₂ and the corresponding JRC values. For comparison purposes, a third synthetic surface was added with JRC = 0 to mimic an idealized parallel plate model (Table 2).

2.2.3. Numerical X-crossing fractures creation

To create an authentic numerical representation of surface roughness in natural rock fractures, encompassing contact areas, asperities, and intricately varying apertures with Gaussian distribution (Pyrak-Nolte, 1999), an anisotropic fracture model was developed. This model accounts for spatial variability in aperture sizes and asperity contacts.

The models' construction involved a numerical uplift step, where the scanned fracture surfaces were elevated by 1.0 mm and 0.7 mm, respectively to mimic the aperture. This was followed by a shear displacement, shifting the upper surface along the x direction by 1.0 mm for all the models. These steps allowed the generation of distinct fracture geometries reflecting realistic geomechanical deformations.

Asperity interlocking and contact zones were captured by combining vertical aperture adjustment with lateral shear displacement applied to

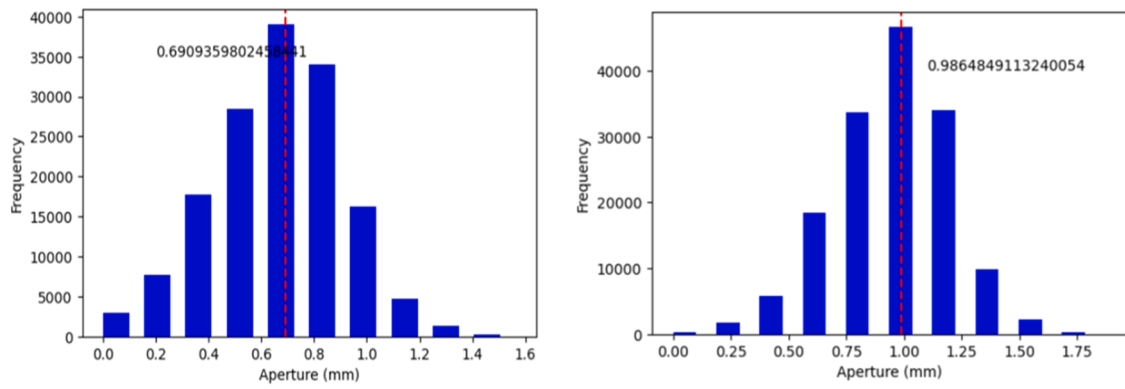


Fig. 4. Aperture distribution for single 10 JRC fractures (0.7 mm left and 1 mm right).

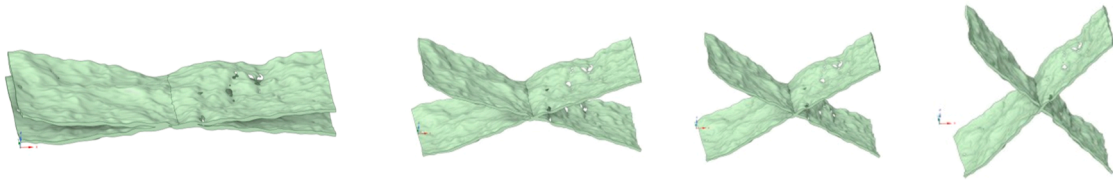


Fig. 5. Examples of 3D X-crossed fracture models JRC=10, 0.7 mm, intersection angle 10°, 40°, 60° and 90°, respectively, (white spots within the green surface represent contact area).

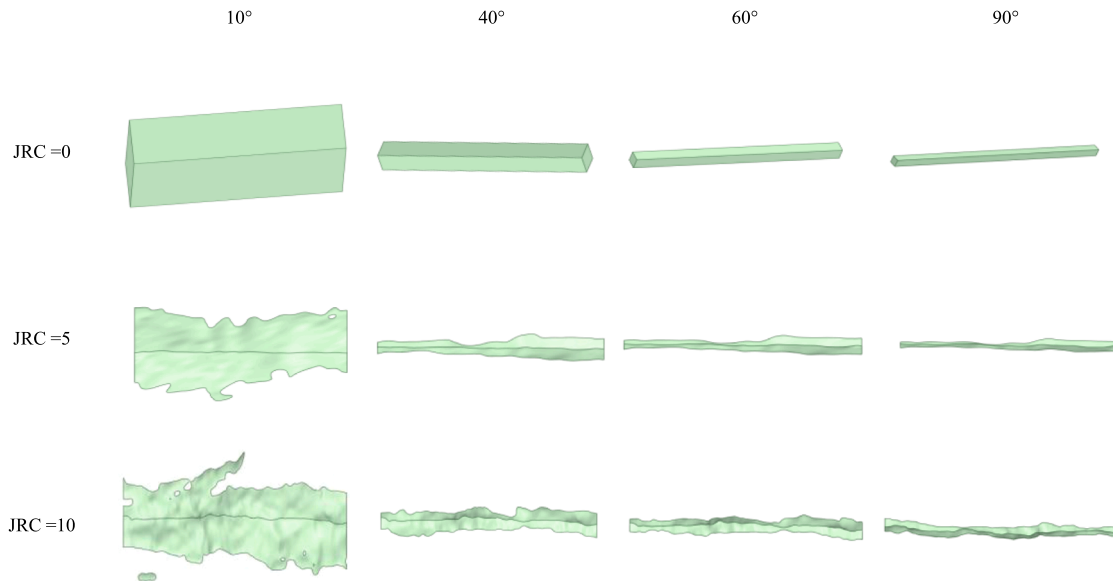


Fig. 6. Three-dimensional perspective of the fracture intersection at different roughness and angles (1 mm aperture models).

one fracture surface. Contact zones emerged naturally from the surface overlap of scanned geometries and were incorporated into the CFD models as impermeable, no-slip wall boundaries. Deformable contact mechanics were not considered. Examples of the aperture distribution for the two single rough fractures are presented in Fig. 4. In this study, the mean aperture was taken as the representative value and directly used in the simulations.

Subsequently, these single fracture models were duplicated and rotated along the y-axis to 10, 40, 60 and 90°, respectively, resulting in the creation of X-shaped fracture models (Fig. 5).

Intersecting fractures introduce junction zones that alter local flow fields through added tortuosity and spatially variable aperture. This intersection, combined with the roughness of the fracture walls, plays a

crucial role in shaping the fluid flow and transport behaviors within the fractures. Fig. 6 provides a 3D visualization of the intersections in three rough-walled models.

In the case of the X-crossed parallel plate fractures, the intersections take the form of flat parallelepipeds that become narrower as the intersection angle increases. However, as the roughness of the fracture walls increases, the intersections become increasingly irregular. This irregularity is characterized by varying aperture sizes, diverse contact areas, and the development of multiple segments, particularly evident in the case of a 10° intersection.

Geometric variations, particularly at fracture intersections, generate spatially heterogeneous velocity fields and pressure gradients. The interaction between surface roughness and intersection geometry

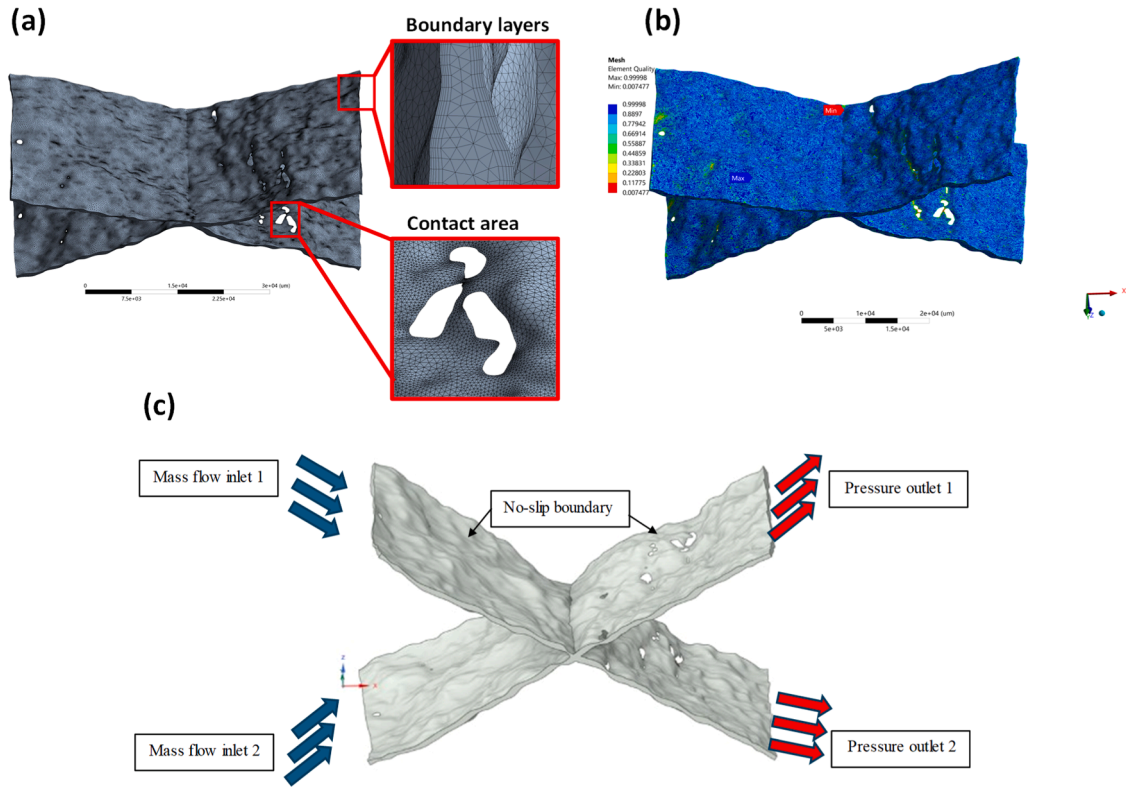


Fig. 7. a) An example of meshing used in numerical simulations b) Example of the mesh quality distribution. c) Schematic of the boundary conditions. Example JRC=10 0.7 mm 60°.

creates irregular aperture distributions and tortuous flow paths, leading to significant variability in fluid transport. As a result, flow behavior in crossed fractures is highly complex and demands detailed analysis to accurately characterize inertial effects and predict hydraulic performance

2.3. Numerical modeling and meshing

2.3.1. Mesh generation

The fracture surfaces were imported into ANSYS Fluent to construct computational domains for Direct Numerical Simulation (DNS) of incompressible Newtonian fluid flow by solving the Navier-Stokes equations (Eq. (8)).

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) + \nabla p = \mu \nabla^2 \mathbf{u} + \mathbf{F} \quad (8)$$

Where the flow velocity vector is given by $\mathbf{u}$, the fluid density by ρ , the pressure gradient of the fluid by ∇p , and the viscosity coefficient by μ .

Mesh generation used a high-quality tetrahedral scheme with refined boundary layers near fracture walls, employing 6–16 million elements depending on roughness and aperture size.

The tetrahedron meshing method was used over the three-dimensional geometric models in order to better fit the fracture replica (Fig. 7a). The complexity and the number of tetrahedron elements depended on the fracture roughness and the aperture.

Due to the intricate flow dynamics induced by roughness, boundary layered mesh was necessary to refine the mesh elements adjacent to the walls, ensuring a more precise flow simulation. This mesh type enhances mesh quality and improves numerical convergence. In addition, finer tetrahedral elements were used at the intersections and contact regions, ensuring a more precise flow simulation

To ensure that the results are independent of mesh resolution for

Table 3

Summary of DNS simulation cases and parameters.

Parameter	Values / Description
Fracture Geometry	3D X-shaped intersections
Intersection Angles (°)	10°, 40°, 60°, 90°
Fracture Roughness (JRC)	0 (Parallel Plate), 5 (Medium), 10 (High)
Apertures (mm)	0.7 mm, 1.0 mm
Number of Models	24 models (3 roughness levels × 2 apertures × 4 angles)
Flow Rates	$Q = 1.67 \times 10^{-9}$ to 3.33×10^{-6} m ³ /s
Reynolds Number Range	Approximately 1 to 100
Boundary Conditions	Inlet: Constant velocity inlet Outlet: Zero-pressure outlet
Working Fluid	Water (Density $\rho = 998.2$ kg/m ³ , Viscosity $\mu = 0.001$ Pa·s)

each model, a substantial number of mesh elements ranging between 6 million and 16 million have been employed, contingent upon the aperture and roughness characteristics.

Two mesh metrics are utilized for evaluating mesh quality. The first is the element quality, ranging from 0 to 1 (see Fig. 7b). The second is the aspect ratio, serving as an indicator of cell stretching.

In this study, we insured that all the prepared models targeted an aspect ratio close to 1 and 95 % of the elements have above 0.6 element quality.

Based on the number of elements used in each model, there is no difference in the calculation results. However, the computational time duration increases. Therefore, the precision of mesh discretization can ensure the accuracy of the calculation results.

2.3.2. Flow conditions and boundary setup

Boundary conditions and the simulation parameters are summarized in Table 3. The outlet was set to a zero-pressure boundary (Fig. 7c).

The software was employed to sequentially solve the Navier-Stokes

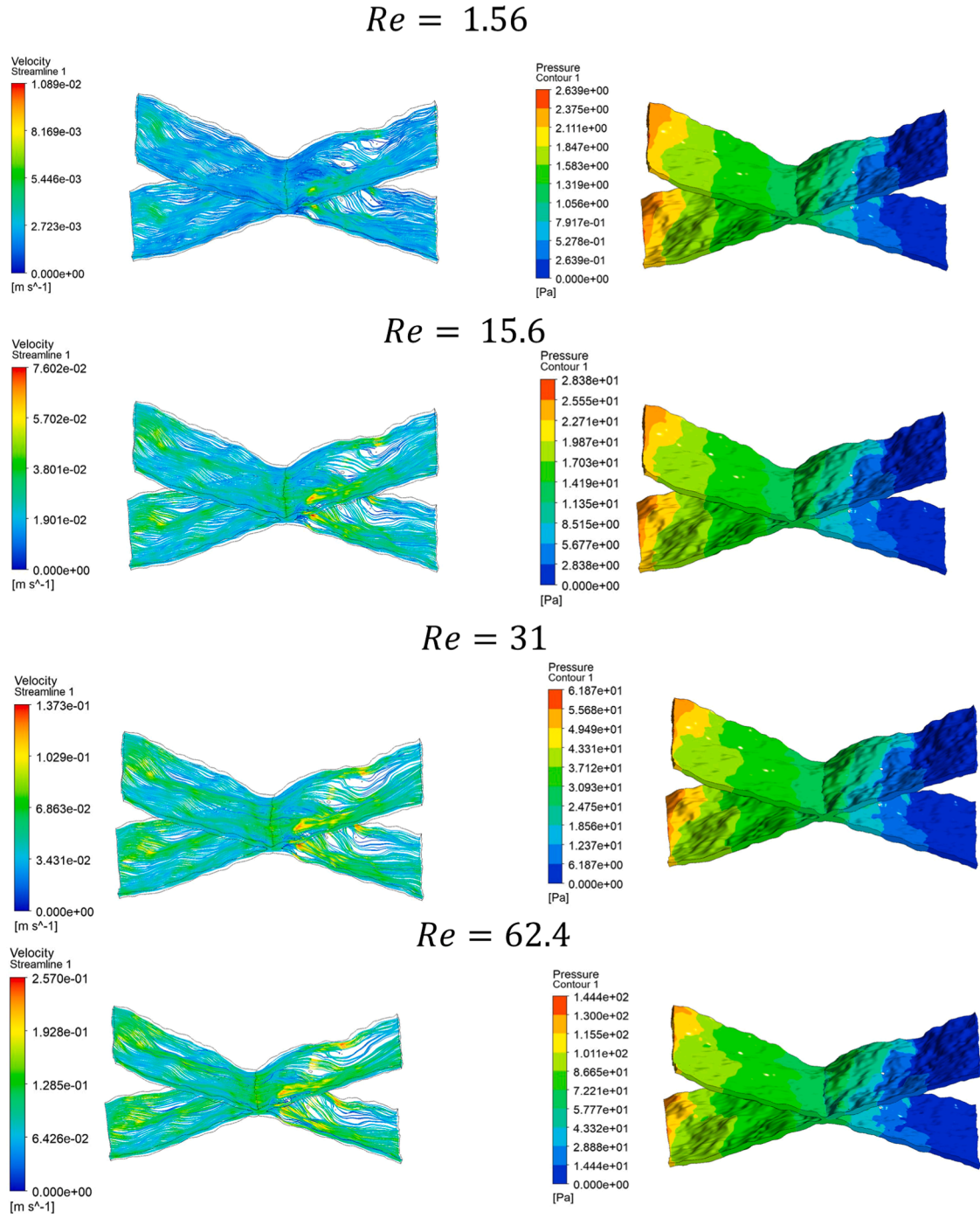


Fig. 8. 3D distribution of (left) streamlines and (right) pressure drop fields for a model with JRC = 10, aperture = 1.0 mm, and intersection angle = 40° at different Reynolds numbers. Flow enters from the left inlet and exits from the right.

Equation (NSE) (Eq. (8)), a commonly utilized equation for describing Newtonian viscous fluid flow at a microscopic scale. These tools have a broad history of application in the analysis of fluid flow through rock fractures, as demonstrated by previous studies (Crandall et al., 2010; Huang et al., 2018; Zimmerman and Yeo, 2000).

We employed residuals as the primary convergence indicators, as they are the most common means of measuring the imbalance in the governing equations, including continuity, momentum, and energy. For laminar flow, we adopted a typical residual threshold of 10^{-10} or lower, which is generally considered acceptable. Convergence in the simulation was assumed to be achieved when the residual values flattened out and exhibited minimal fluctuation between successive iterations.

All simulations were conducted on a High-Performance Computer (HPC) with 128 GB of memory allocated per simulation. A mesh independence analysis indicated that the chosen mesh number adequately represents the fracture domain, as additional mesh refinement had a minimal impact on the results while significantly increasing computation costs.

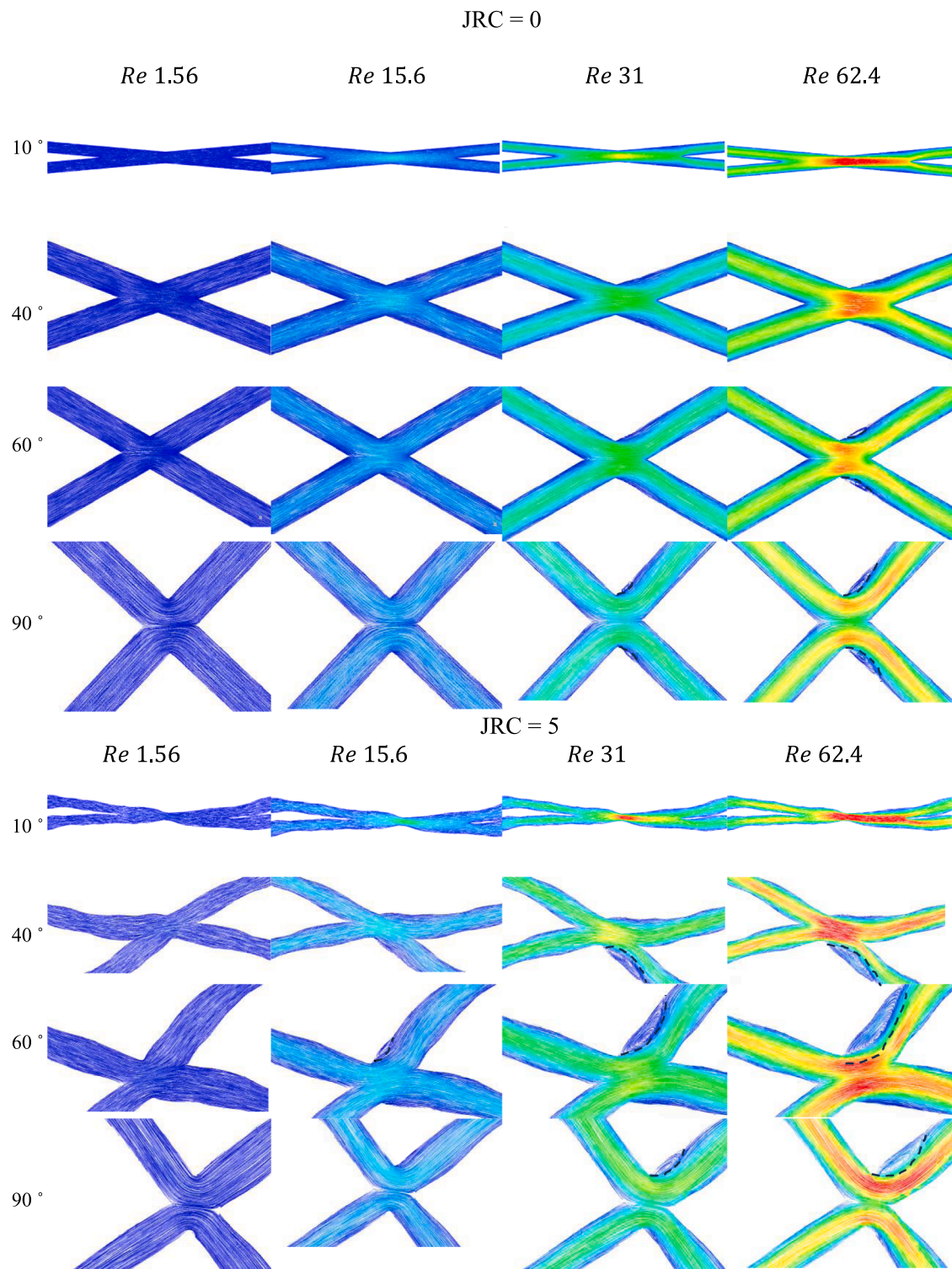


Fig. 9. 2D Flow streamlines in the fracture intersections over a range of Reynolds number from 1.56 to 62.4 and varying Joint Roughness Coefficients (JRC): 0 JRC, 5 JRC, and 10 JRC, illustrating the formation of eddies at the fracture intersections, across intersection angles of 10°, 40°, 60°, and 90°. All models are at an aperture of 0.7 mm. Flow is from left to right.

3. Results

3.1. Fracture flow

3.1.1. General flow patterns

The streamlines in the parallel plate models (JRC = 0) are uniform

and symmetrical no matter the intersection angle, and the fluid coming from inlet-1 is exclusively directed towards outlet-1 (similarly for inlet-2 and outlet-2). However, with an increase in roughness and a decrease in aperture, some streamlines originating at inlet-1 may be diverted to outlet-2, while simultaneously some streamlines originating at inlet-2 may be diverted to outlet-1. Low-angle intersections frequently exhibi-

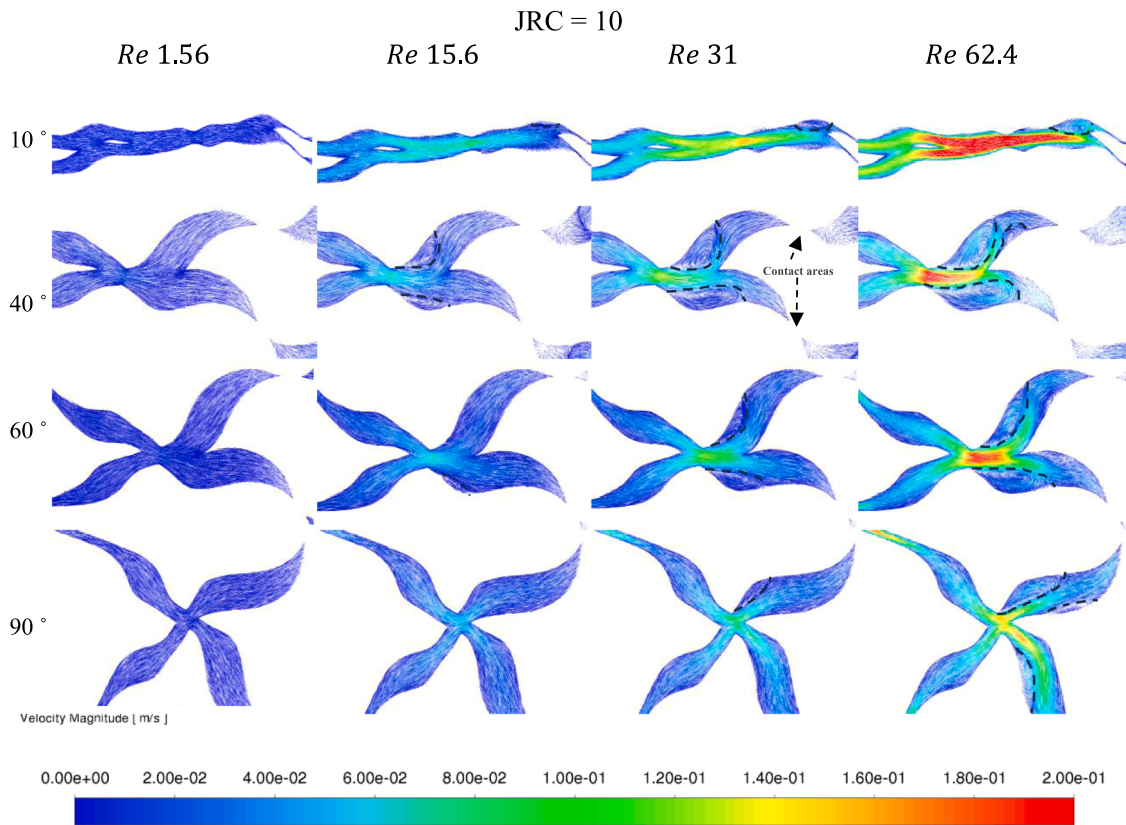


Fig. 9. (continued).

ted increased contact regions and narrowed apertures, which likely intensified flow resistance and pressure drops. As a result, the behavior of streamlines and pressure drop in these areas may differ significantly from that of high angle intersection models. 3D distribution of streamlines and pressure drop through JRC = 10, 1 mm and 40° model is shown in Fig. 8.

3.1.2. Viscous eddies at the intersections

Fig. 9 illustrates 2D streamlines at the intersections at an aperture of 0.7 mm, depicting the magnitude of velocity and capturing the flow patterns for various Reynolds numbers (Re). These visualizations highlight a clear transition from laminar (Darcy) flow to inertial (non-Darcy) flow regimes, accompanied by the emergence and growth of eddies at the intersections.

At $JRC = 0$, the fractures are smooth, resulting in symmetric and undisturbed streamline profiles at low Reynolds numbers. At $Re = 1.56$, there are no visible eddies regardless of the intersection angle. As Re increases, eddies begin to appear selectively at larger angles. At $Re = 15.6$, small eddies emerge at 60° and 90° intersections, while 10° and 40° remain largely streamlined. At the highest Reynolds number ($Re = 62.4$), the eddies become more distinct in these wide-angle intersections, but still relatively few in number and small in size due to the smooth walls offering little resistance or disturbance to the flow.

In contrast, $JRC = 5$ introduces moderate roughness, which accelerates the onset of eddy formation. Even at $Re = 15.6$, small vortices begin to appear at 40° and become more pronounced at 60° and 90°. At $Re = 31.2$ and above, the number of eddies significantly increases, especially in wider intersections, and their sizes become larger than those observed in smooth fractures. The roughness enhances local turbulence, creating asymmetric flow separation and multiple small-scale vortices, especially at sharp corners or where flow is redirected.

With $JRC = 10$, the flow becomes highly perturbed even at low Reynolds numbers. While $Re = 1.56$ still shows laminar-like profiles,

there are minor flow deviations and recirculation hints, particularly at 60° and 90°. As Re rises to 15.6 and 31.2, the number of eddies increases dramatically. Eddies form in almost all angles except the narrowest (10°). At $Re = 62.4$, the intersections at 40°, 60°, and 90° are dominated by large, fully developed vortices that span most of the junction area. These vortices are not only more numerous but also significantly larger compared to those in lower JRC cases. In some frames, 2–3 major eddies can be identified per intersection, and their diameters are comparable to the width of the branches themselves, indicating strong inertial effects.

Fig. 10 illustrates the hydraulic conductivity (K) responds to increasing Re , under varying JRCs, and different fracture apertures, allowing a comprehensive analysis of how geometry and roughness influence fluid transport through fracture intersections.

Across all conditions K remains nearly constant at low Reynolds numbers, corresponding to the Darcy (viscous-dominated) regime, and begins to decline as Re increases, signaling the transition into non-Darcy (inertial-dominated) flow. This transition is relatively mild for smooth fractures ($JRC = 0$), especially at narrow intersection angles such as 10° and 40°, where streamline curvature is minimal and energy loss is low. In these cases, hydraulic conductivity is relatively stable even up to $Re \approx 100$, particularly for the larger aperture (1 mm).

As fracture roughness increases (from $JRC = 0$ to $JRC = 10$), the impact of Reynolds number on hydraulic conductivity becomes more pronounced. At $JRC = 10$, even moderate increases in Re result in a steep decline in K , particularly at wider intersection angles (60° and 90°). This reflects enhanced inertial losses due to flow separation, eddy formation, and stagnation zones at the rougher fracture surfaces.

Aperture plays an important role as well. In general, fractures with larger apertures (1 mm) exhibit higher absolute values of K compared to those with smaller apertures (0.7 mm), across all JRC and angle combinations. However, this increase in aperture does not eliminate the non-Darcy effects observed at high Re . Even for the 1 mm aperture with high roughness ($JRC = 10$), the conductivity still drops markedly at larger

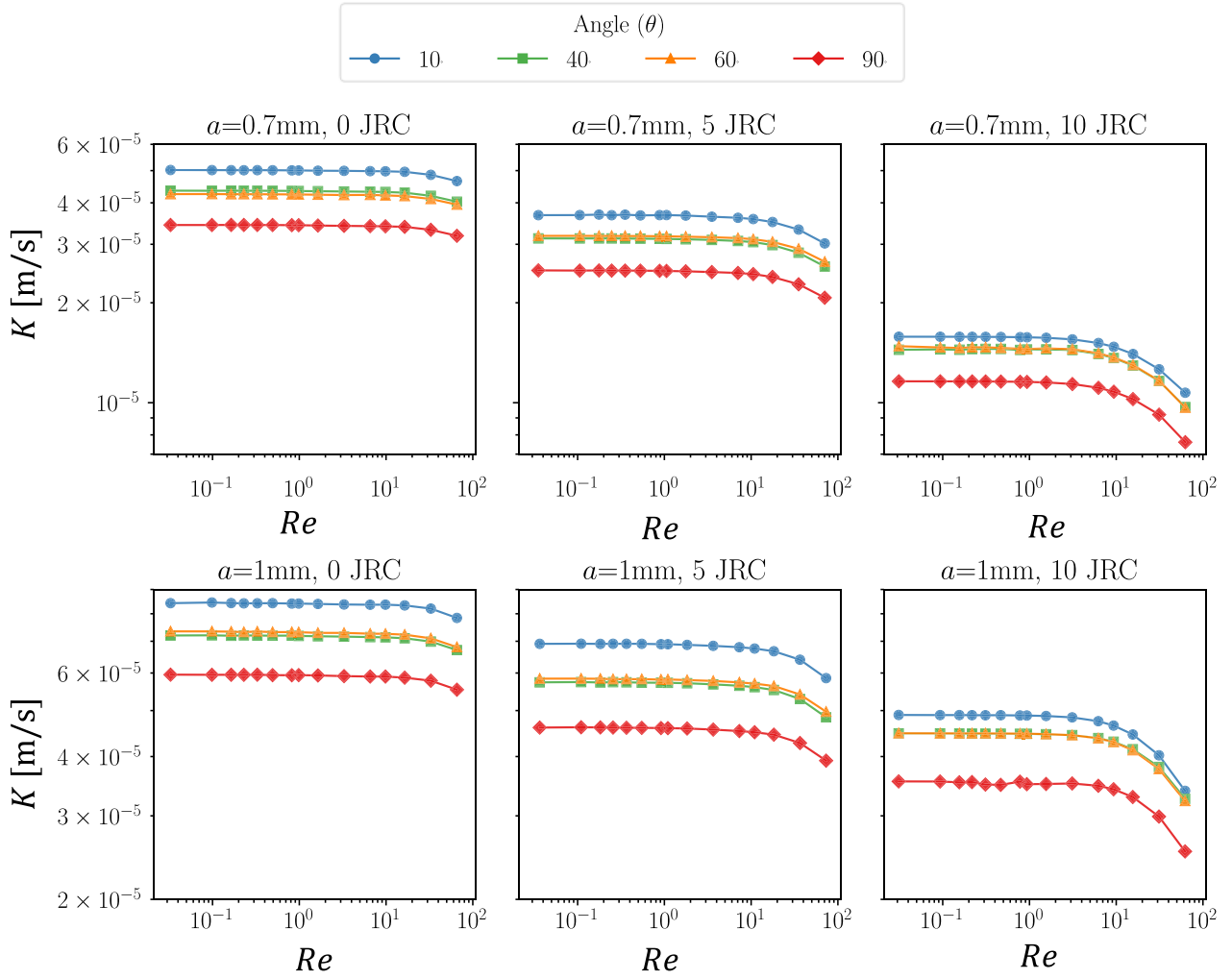


Fig. 10. Effect of JRC, aperture, and θ on K Across Re .

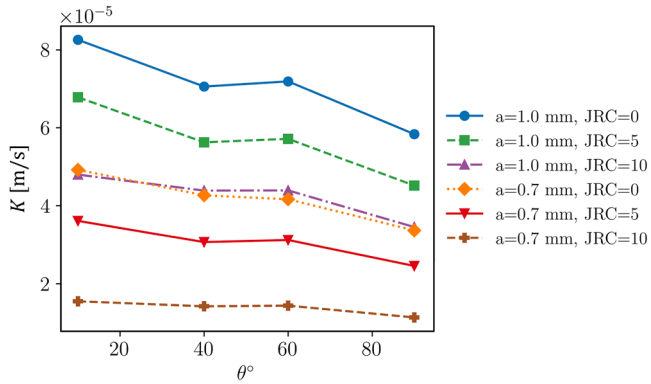


Fig. 11. Relationship between hydraulic conductivity K and intersection angle θ at different JRCs and apertures.

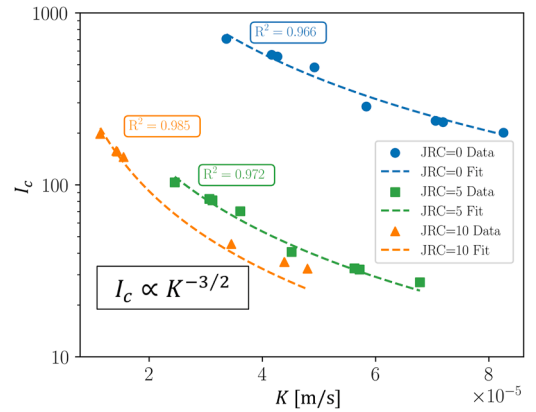


Fig. 12. The critical head gradient I_c dependence on K .

angles as Re increases. This indicates that while aperture facilitates higher base flow rates, it cannot fully mitigate the inertial losses introduced by fracture geometry and roughness.

To isolate the role of intersection angle on hydraulic conductivity, we analyzed the variation of K with respect to θ across all JRC levels (Fig. 11). The results indicate that K in general decreases as θ increases, with this trend being more significant for smoother fractures (lower JRC values) and larger apertures. The highest K values are observed in the

case of $a = 1.0$ mm and $JRC = 0$, while the lowest values occur for $a = 0.7$ mm and $JRC = 10$. While the general trend indicates that K decreases as θ increases, the rate of this decrease is not uniform across all angles. For most cases, hydraulic conductivity exhibits a relatively stable or slightly decreasing trend between intersection angles of 40° and 60° , before resuming a more pronounced decline at larger angles. This suggests a reduced sensitivity of K to changes in intersection angle within the tested specific range. However, the overarching effect of increasing

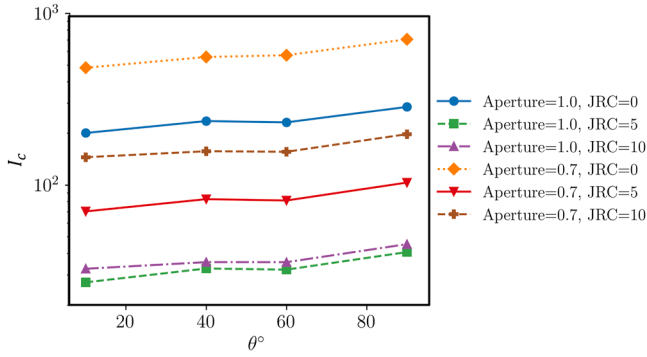


Fig. 13. Relationships between the critical hydraulic gradient I_c and intersection angles θ .

intersection angle is a reduction in hydraulic conductivity due to enhanced tortuosity and flow resistance.

Under Darcy's law, (K) remains constant but starts to decline as flow enters the non-Darcy regime at higher (i) . The threshold gradient marking the onset of non-Darcy flow is referred to as the critical hydraulic gradient (I_c). This value is identified when the reduction in K exceeds 5 %, after which K continues to decrease as i increases further.

Fig. 12 depicts the determined I_c for all the simulations. We find that I_c follows an inverse power law with, $I_c \propto K^{-3/2}$, as demonstrated by the fitted curves for different JRC values with $R^2 \geq 0.966$, which is consistent with the literature (Singh et al., 2024; Wang et al., 2022). This inverse power law indicates that as hydraulic conductivity increases, a smaller hydraulic gradient is needed for the transition to non-Darcy flow, whereas lower conductivity would require a higher gradient.

The relationships between I_c and θ are shown in Fig. 13, I_c increases with the increasing θ for different fracture apertures and JRCs. This behavior indicates that larger intersection angles require more force for inertia to initiate.

The relationship between JRC and I_c was non-monotonic, with I_c decreasing from JRC = 0 to 5, then increasing at JRC = 10. This suggests competing effects of turbulence onset and energy dissipation. This suggests that moderate surface roughness (JRC = 5) promotes earlier onset of nonlinear flow, likely due to enhanced turbulence and eddy formation that destabilize the flow regime at lower gradients. In contrast, highly rough fractures (JRC = 10) increase energy dissipation and flow resistance, thereby requiring higher hydraulic gradients to initiate nonlinear behavior.

The results also show that JRC = 0 (representing a smooth, parallel plate scenario) produces higher I_c values than moderate roughness. This outcome is consistent with the idea that, in the absence of surface asperities, flow remains more streamlined and laminar, delaying the transition to nonlinearity. Without roughness-induced flow disturbances, such as eddy formation or recirculation zones, higher gradients are necessary to trigger nonlinear flow.

In terms of fracture geometry, increasing the intersection angle is θ associated with a gradual rise in I_c . This trend holds across all JRC and aperture values. Larger intersection angles introduce greater tortuosity and more complex flow paths, which act as constrictions. As a result, more energy is required to overcome this added resistance, leading to elevated I_c values. The effect of intersection angles reinforces the role of geometrical complexity in influencing nonlinear flow thresholds.

3.2. Darcy to non-Darcy relationships

3.2.1. Forchheimer equation

Understanding the transition from Darcy to non-Darcy flow is essential for accurately modeling fluid movement through complex porous and fractured media. While Darcy's law governs flow under low Reynolds number conditions, it fails to capture the nonlinear behaviors

that arise due to inertial effects as flow rates increase. These deviations from linearity are especially pronounced in rough-walled fractures with irregular geometries, where eddy formation and velocity profile distortion become significant.

Studies such as (Chaudhary et al., 2011, 2013; Xing et al., 2025; Zhou et al., 2020) demonstrated that the transition from Darcy to non-Darcy flow in diverging-converging pore geometries is driven primarily by the growth of eddies within pores as Reynolds number increases. These eddies reduce the effective flow area and create additional resistance, which manifests as a drop in the K . The flow transitions through distinct regimes, with Darcy behavior prevailing at low Re , followed by a nonlinear region, and in some cases a reversion to quasi-linear behavior at very high Re as eddy growth saturates.

The onset of non-Darcy behavior is best described using the Forchheimer equation that extends Darcy's law by incorporating a quadratic term that accounts for inertial losses. Forchheimer model successfully captures both the linear and inertial regimes, demonstrating its broader applicability for describing flow in intersecting fractures (Fig. 14).

3.2.2. Izbash equation

In contrast, the Izbash law is a power-law model suitable for describing flow behavior strictly in the non-Darcy regime. Fig. 15 illustrates this limitation, where a single power-law fit using Izbash's equation does not adequately represent flow in the Darcy regime, leading to higher residuals when applied across the full dataset.

3.2.3. Fractures characteristics, exponents and coefficients

In the Forchheimer Eq. (2), β represents the coefficient of inertial flow resistance. This coefficient is determined from the parameter b by fitting a quadratic equation to simulation data, where $\beta = bg$. A higher observed β with decreasing K highlights the growing significance of inertial effects. A larger β value directly indicates stronger inertial resistance within the flow regime, emphasizing the dominance of non-linear behavior as the flow transitions from Darcy to non-Darcy conditions.

We find that β increases with surface roughness, with higher values observed for larger JRC values. However, for a given JRC, β follows an inverse power-law relationship with the hydraulic conductivity (Fig. 16a), given by:

$$\beta \propto K^{-\frac{2}{3}} \quad (9)$$

with R^2 of 0.93, 0.95, and 0.77 for JRC = 10, 5, and 0, respectively.

Notable, Eq. (9) is consistent with the independently developed equation by (Zhou et al., 2019) for porous media:

$$K \propto \frac{1}{\beta^{\frac{3}{2}}} \quad (10)$$

When combining β data from all JRC values, we found an alternative power-law relationship (Fig. 16b), where the best fit suggests $\beta \propto K^{-5/3}$ with $R^2 = 0.89$. This indicates, while β decreases with increasing hydraulic conductivity, a higher exponent of 5/3 relative to 3/2 (Zhou et al., 2019) could result from combining β data from a diverse geometry of media classes, such as different JRC fractures in this study.

Fig. 17 presents the variation of β as a function of fracture intersection angle θ for two fracture apertures and JRC. A prominent trend observed in the plot is the strong influence of fracture aperture on β . For all JRC values, the smaller aperture exhibits consistently higher inertial coefficients than the larger aperture ($a = 1.0$ mm). This result highlights the enhanced inertial losses in narrower fractures, where the flow experiences greater confinement, steeper velocity gradients, and stronger interaction with rough surfaces.

Roughness also plays a significant role in modulating inertial effects. At a fixed aperture, increasing JRC leads to a substantial rise in β ,

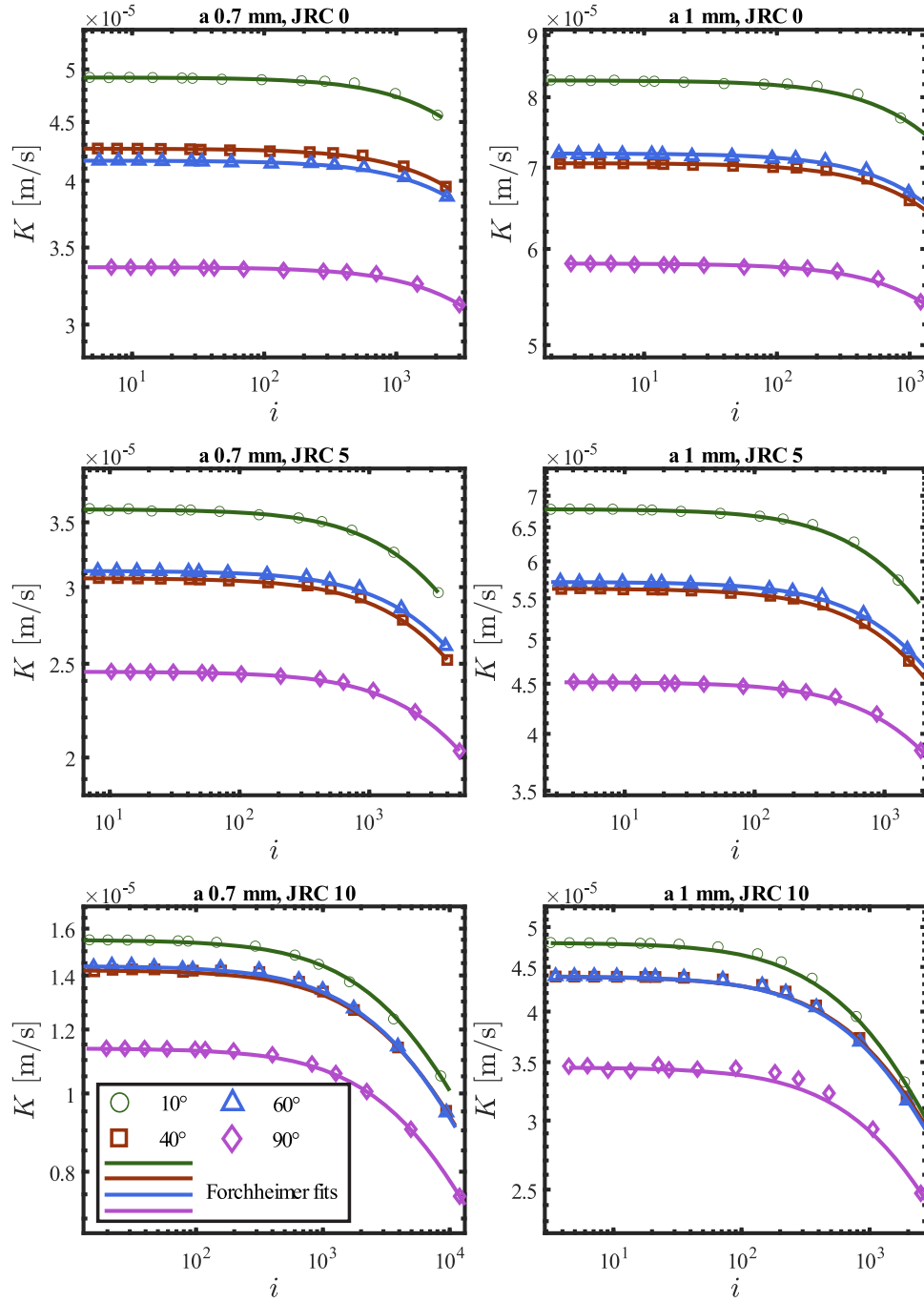


Fig. 14. Forchheimer fitting of hydraulic connectivity K for different apertures, JRC, intersection angles θ and apertures at varying hydraulic gradients i .

indicating stronger flow resistance in rougher fractures. For both apertures, the progression from JRC = 0 to JRC = 10 reflects a clear and monotonic increase in inertial losses, underscoring the importance of surface roughness in non-Darcy flow regimes. This behavior aligns with previous theoretical and experimental studies on rough-walled channels and fractures (Z. Li et al., 2021; Liu et al., 2021).

The influence of fracture intersection angle θ appears more nuanced. While some variation in β is noted across angles ranging from 20° to 90°, the overall effect is less pronounced than that of aperture and roughness. In rougher fractures, there is a mild upward trend of β with increasing angle, suggesting that sharper intersections may contribute slightly to inertial dissipation. However, the consistency of the curves across angles indicates that the dominant controls on β remain aperture and surface roughness rather than intersection geometry.

The Izbash equation's inertial exponent (n) is a key parameter in characterizing the non-linearity of non-Darcy flow behavior. The values of n were determined for different fracture intersection angles (θ) and two different fracture apertures ($a = 1.0$ mm and $a = 0.7$ mm), as shown in Fig. 18.

The results indicate a general inverse relationship between n and K , following the power law $n \propto K^{-0.25}$ and $n \propto K^{-0.15}$ for $a = 1$ and $a = 0.7$ respectively. The exponent n increases with decreasing K demonstrating stronger degree of non-linearity in the flow. Higher n values are associated with more turbulent or inertial-dominated flow conditions, whereas lower n values (closer to 1) reflect near-laminar or Darcy flow. Therefore, in high-permeability cases, such as smoother fractures or those with low roughness (low JRC) and low intersection angles, the flow tends to remain laminar, resulting in lower n values. Conversely,

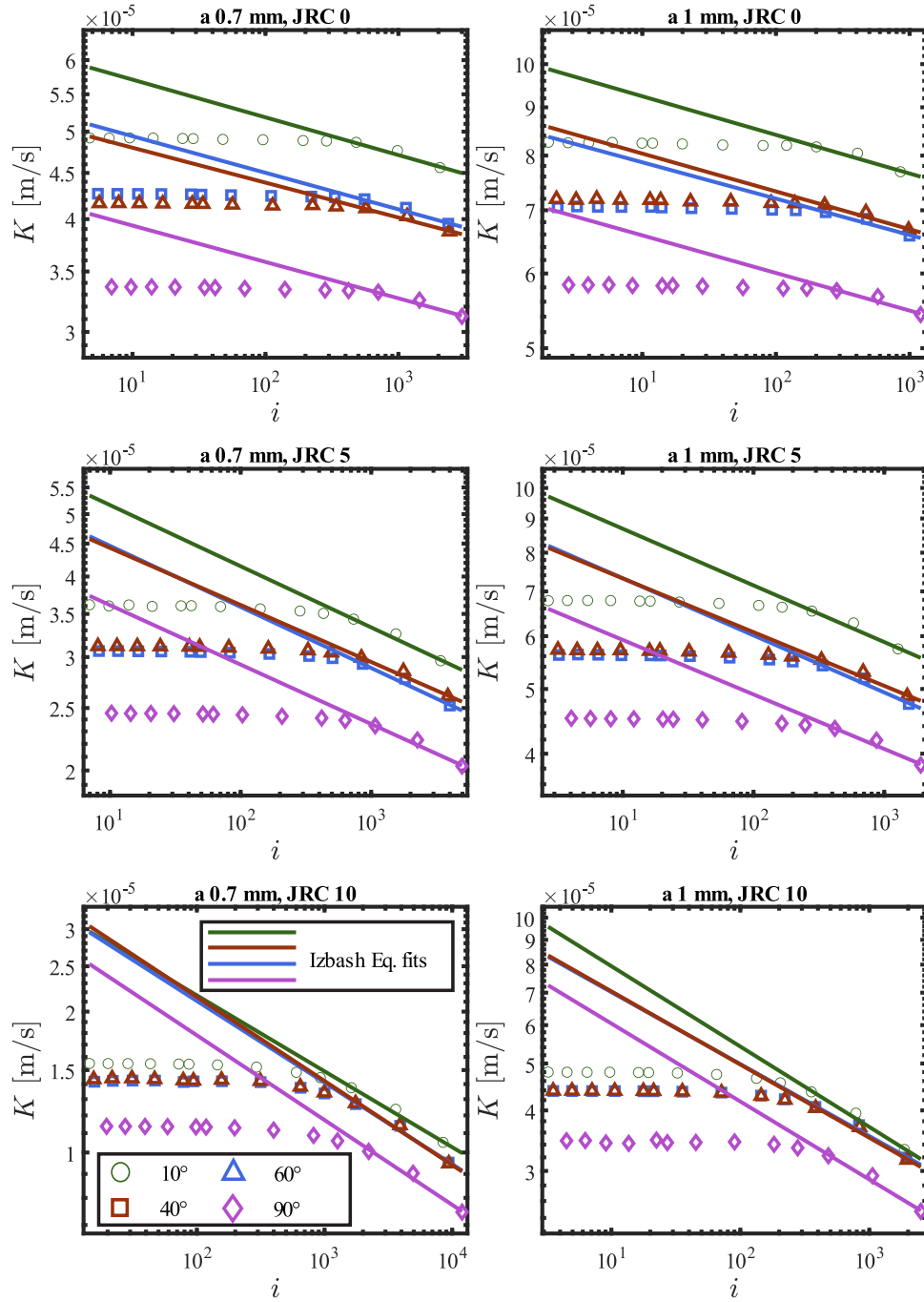


Fig. 15. Izbash fitting results for non-linear zones in fracture intersections at different JRCs, angles and apertures at varying hydraulic gradients i .

lower permeability fractures promote earlier onset of inertial effects, leading to higher n values.

4. Discussion

This study provides new insights into inertial flow behavior in 3D intersecting rough fractures by systematically varying surface roughness, intersection angle, and aperture. While previous research has addressed non-linear flow in simplified or 2D fracture geometries, our work uniquely combines high-resolution scanned rock surfaces with fully 3D DNS to quantify how geometric complexity governs the transition from Darcy to non-Darcy flow. The identification of inverse power-law relationships between critical gradient, inertial coefficients (β , n), and hydraulic conductivity across a wide parameter space represents a novel contribution.

resents a novel contribution.

The results indicate that increasing fracture surface roughness accelerates the transition from Darcy to non-Darcy flow, as evidenced by a decrease in the critical hydraulic gradient (I_c) (Fig. 12). The observed inverse power-law relationship between (I_c) and hydraulic conductivity (K) is consistent with previous studies (Singh et al., 2024), which demonstrated that lower-permeability media require higher forcing to initiate inertial effects. For example, in fractures with $JRC = 10$, I_c decreased by up to 30 % compared to smooth-walled configurations.

This behavior can be attributed to the generation of localized eddies at surface asperities, which promote early momentum dissipation and streamline separation. Fig. 9 illustrates the progressive development of vortices with increasing Reynolds number in fractures of varying JRC, confirming that surface roughness significantly enhances inertial flow

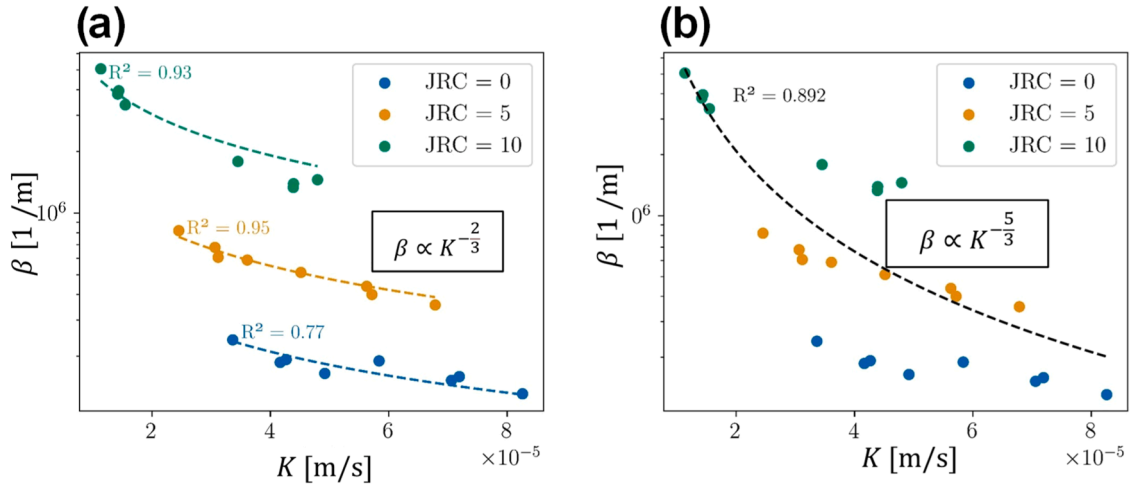


Fig. 16. a) Forchheimer Equation coefficient β relationship with hydraulic conductivity K b) fitting based on JRC b) global fitting.

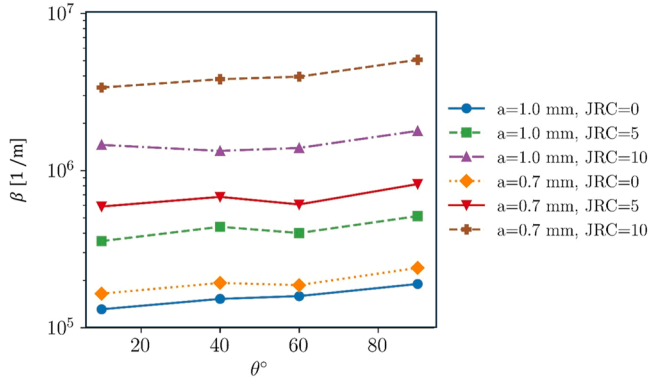


Fig. 17. Forchheimer Equation coefficient β versus intersection angle θ .

resistance. This is further corroborated by the increase in the Forchheimer coefficient (β) and the Izbash exponent (n) with increasing JRC (Figs. 16 and 18), consistent with eddy-driven flow distortion and turbulence reported in previous numerical and experimental investigations (Xing et al., 2025; Zhou et al., 2020).

The intersection angle (θ) also modulates the onset and severity of non-Darcy behavior. Simulations reveal that wider intersection angles (e.g., 90°) exhibit delayed onset of nonlinearity, as indicated by elevated

I_c values (Fig. 13). This response is attributable to the spatial redistribution of inertial forces across a broader junction zone, reducing localized velocity gradients. However, at higher Reynolds numbers, such intersections exhibit larger and more stable eddies, as demonstrated in Fig. 9. These flow features correspond to the development of symmetric counter-rotating vortices, which dominate the intersection region under rough-walled conditions (JRC = 10).

The findings are in agreement with Kim and Yeo, 2025 who observed similar vortex structures in rough-walled intersection experiments. Additionally, (Liu et al., 2022) reported that increasing intersection angles enhance flow separation and head loss, a trend confirmed by our DNS results, particularly under high JRC conditions. Collectively, these observations underscore the role of intersection geometry in controlling inertial dissipation pathways.

A principal contribution of this study lies in the quantitative resolution of microscale flow features within realistic 3D geometries obtained from high-resolution rock surface scans. Unlike previous 2D models or idealized fracture geometries, the present DNS approach captures eddy dynamics at asperities and quantifies their influence on flow resistance metrics.

The power-law relationships derived for β , I_c , and n extend the formulations of Zhou et al. 2019 by integrating explicit geometric parameters (JRC, θ , aperture) into predictive models. These relationships provide a mechanistic basis for improved representation of non-Darcy flow in fractured media and support the development of more

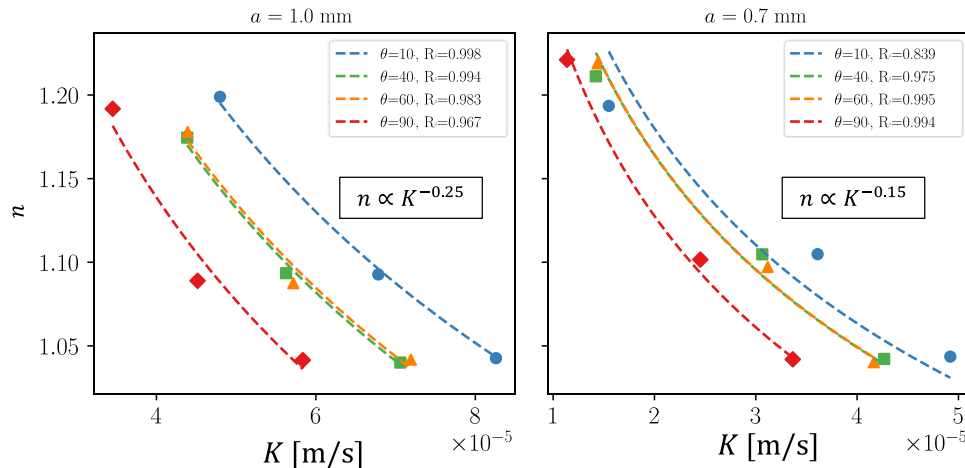


Fig. 18. Relationship between the exponent n from the Izbash equation and hydraulic conductivity K across fractures with varying apertures, and JRC.

Table 4

Comparison of power-law relationships developed in this study with previous models.

Study	Power-Law Relation	Remarks
This study	$\beta \propto K^{-\frac{5}{3}}$ $I_c \propto K^{-2/3}$	Includes fracture roughness (JRC), intersection angle (θ), and aperture (a). Developed from 3D DNS of X-intersections using real scanned rock surfaces.
Zhou et al. (2019)	$\beta \propto K^{-3/2}$	Derived for porous media, without explicit geometric parameters. Applies to general geologic materials.
Z. Wang et al. (2022)	$I_c \propto s_1 a^{m_1}$ $\beta \propto s_2 a^{m_2}$ (s_1, s_2, m_1 , and m_2 are fitting parameters)	X-crossed 2D intersections with limited roughness variation.
Liu et al. (2021)	$\beta \propto \frac{1}{w^{4-l} e_h^l}$ w is the fracture width e_h is the hydraulic aperture l is an empirical exponent	Accounts for some geometric effects, but based on single rough fracture
Zhang et al. (2022)	$\beta = 120.4 \left(\frac{e}{\sigma_E} \right)^{-1.3}$	The derived power-law is based on single rough fracture

accurate upscaling methodologies for field-scale simulations. Table 4 summarizes the comparison of power-law relationships developed in this study with previous models.

The empirical trends established herein are applicable to a wide range of subsurface engineering scenarios where accurate prediction of flow behavior in fractured rock is critical. The development of a unified quantitative expression that accounts for the combined influence of aperture, roughness, and intersection angle remains a compelling direction for future work. This would enable general predictive modeling of non-Darcy flow across diverse fractured systems and could benefit from scaling analysis or machine learning-based multivariate regression trained on extended simulation datasets.

In particular, the demonstrated dependence of nonlinearity on roughness and intersection geometry highlights the necessity of including these parameters in flow models to avoid underestimation of pressure drops and mischaracterization of fracture transmissivity in the inertial regime.

Several limitations merit consideration. The simulations were restricted to single-phase, Newtonian fluids under isothermal conditions. In practice, subsurface systems may involve multiphase flow and thermal-mechanical coupling, which could alter aperture distributions and flow resistance. For instance, the exclusion of Thermo-Hydro-Mechanical (THM) effects may lead to underestimation of aperture closure or dilation under stress variations.

While the selected apertures were sufficient to reveal clear trends in hydraulic behavior under different roughness and intersection angles, the relatively narrow aperture range limits the extrapolation of these findings to a wider spectrum of fracture openings. Future studies should explore a broader range of aperture sizes, particularly to investigate scaling effects and nonlinear interactions between aperture, roughness, and inertial flow regimes.

Finally, while the DNS approach provides high-fidelity insights, experimental validation is needed to support the derived coefficients and flow thresholds. Integration with laboratory experiments, particularly using 3D-printed rough fracture replicas, would enhance confidence in the applicability of the presented relationships to real systems.

The findings in this study provide important insights into inertial flow behavior in X-shaped fracture intersections with symmetric boundary conditions, this represents a specific configuration within a broader range of possible fracture geometries and loading scenarios. Other intersection types, such as Y- or Z-shaped geometries, and non-symmetric boundary conditions, may exhibit different inertial flow behavior and may require different scaling considerations. Extending

this work to such cases is therefore a key direction for future research. We have chosen to focus on this controlled geometry to allow for systematic analysis and to establish a foundation for broader generalization.

5. Conclusions

This study employed three-dimensional Direct Numerical Simulations (DNS) to investigate inertial flow phenomena in intersecting rough-walled fractures. High-resolution fracture models were generated using scanned natural rock surfaces with controlled roughness, aperture, and intersection angles. The results provide quantitative insights into how fracture geometry and flow conditions influence the transition from Darcy to non-Darcy regimes and the associated hydraulic behavior.

Fracture surface roughness was identified as the primary driver of inertial flow behavior. Increasing roughness, measured via the Joint Roughness Coefficient (JRC), significantly lowers the threshold for inertial onset, promotes eddy formation, and increases pressure losses. Intersection angle was shown to amplify these effects, with wider angles (60°–90°) enhancing streamline deflection and turbulence. While increasing aperture generally improves baseline hydraulic conductivity, it does not suppress the onset of inertial losses in geometrically complex or highly rough fractures.

The transition to non-Darcy flow was characterized using both the Forchheimer and Izbash models. The critical hydraulic gradient I_c was shown to follow an inverse power-law relationship with hydraulic conductivity $\beta \propto K^{-\frac{2}{3}}$. Additionally, The Izbash exponent n also varied systematically with fracture roughness and geometry ranging from 1.2 to 1.8. These empirical relationships provide a foundation for modeling permeability and flow resistance in fractured media under non-linear flow conditions.

Importantly, these correlations were derived under laminar flow regimes with symmetric inlet and outlet boundary conditions, ensuring stable and controlled flow development across fracture geometries. The derived power-law correlations between β , I_c , n and hydraulic conductivity offer a foundation for improving permeability models in fractured geothermal systems. The reduction in hydraulic conductivity at higher Reynolds numbers implies that energy extraction models should incorporate inertial losses due to rough fracture surfaces. Similarly, in CO₂ sequestration, the relationships provide a basis for adjusting injectivity models under high-pressure conditions, accounting for eddy formation and flow separation at fracture intersections. In hydraulic fracturing and mine dewatering, the critical gradient behavior aids in estimating when inertial effects begin to dominate flow, which is crucial for predicting fluid transport and designing efficient drainage or stimulation strategies.

This work enhances the understanding of non-Darcy flow regimes in complex fracture networks and provides transferable tools for incorporating geometric and inertial effects into subsurface flow models. These findings are directly applicable to the design and simulation of energy, environmental, and geotechnical systems involving flow through fractured rocks.

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CRediT authorship contribution statement

Ghoulem Ifrene: Writing – original draft, Methodology, Formal analysis. **Kuldeep Singh:** Writing – review & editing, Validation, Supervision, Methodology, Data curation. **Richard A. Schultz:** Writing – review & editing. **Prasad Pothana:** Visualization, Software. **Neal Nagel:** Writing – review & editing, Supervision, Investigation. **Sven Egenhoff:** Writing – review & editing, Writing – original draft,

Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

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Table 1
Key variables and symbols.

Symbol	Description
q	Darcy flux (m/s)
i	Head gradient (-)
β	Forchheimer inertial coefficient (m ⁻¹)
g	Acceleration due to gravity (m/s ²)
n	Power-law exponent in the Izbash equation
Re	Reynolds number (-)
ρ	Density of water (kg/m ³)
μ	Viscosity of water (Pa·s or N·s/m ²)
I_c	Critical hydraulic gradient (-)
K	Hydraulic conductivity (m/s)
Q	Volumetric flux (m ³ /s)
a	Aperture
α	Viscos coefficient ($\alpha=1/K$)
θ	Angle
v	Average fluid velocity (m/s)
k_v	Viscous permeability (m ²)
k_i	Inertial permeability (m ²)
$-\nabla p$	pressure gradient
F	Body forces per unit volume

Data availability

Data will be made available on request.

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